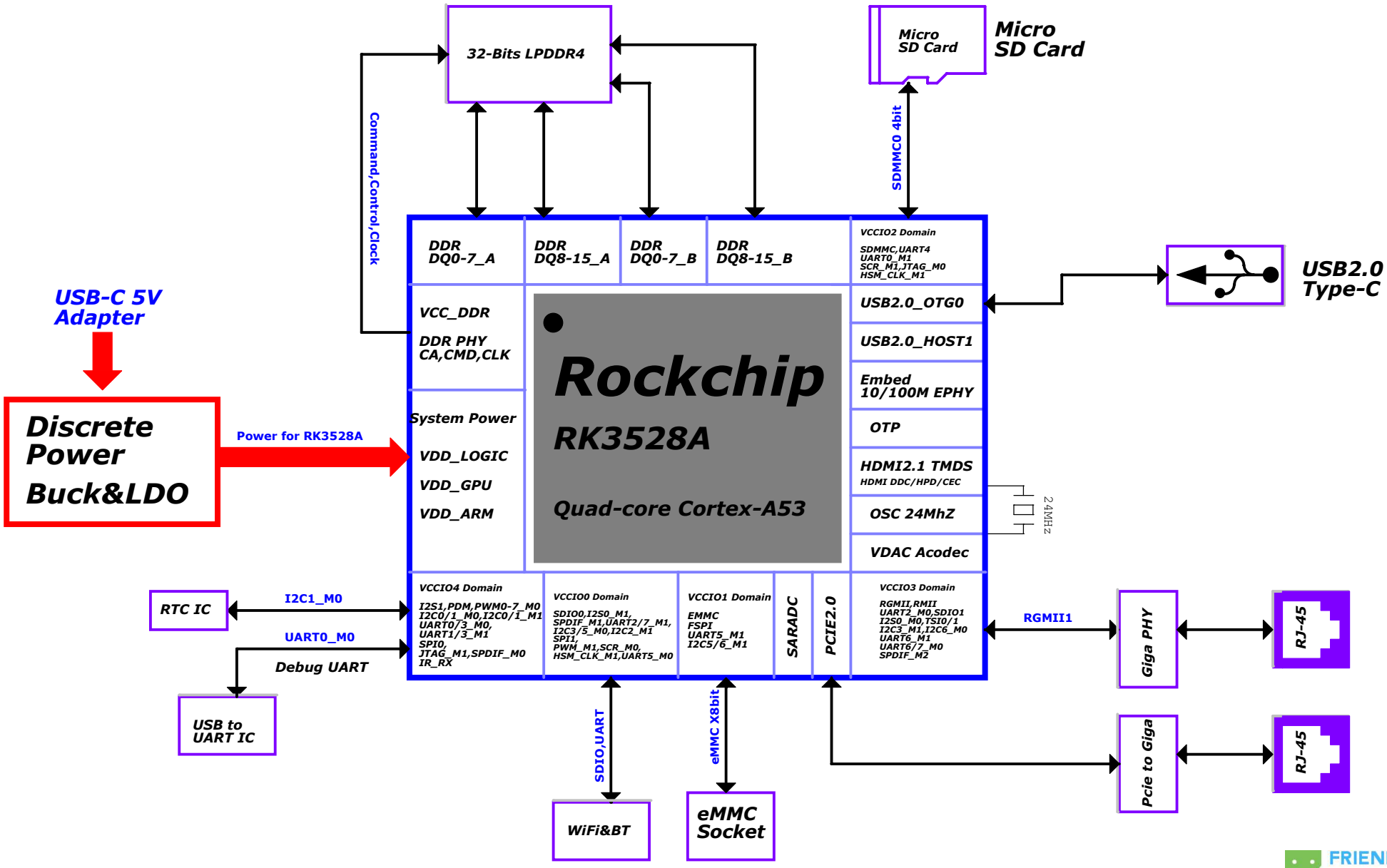
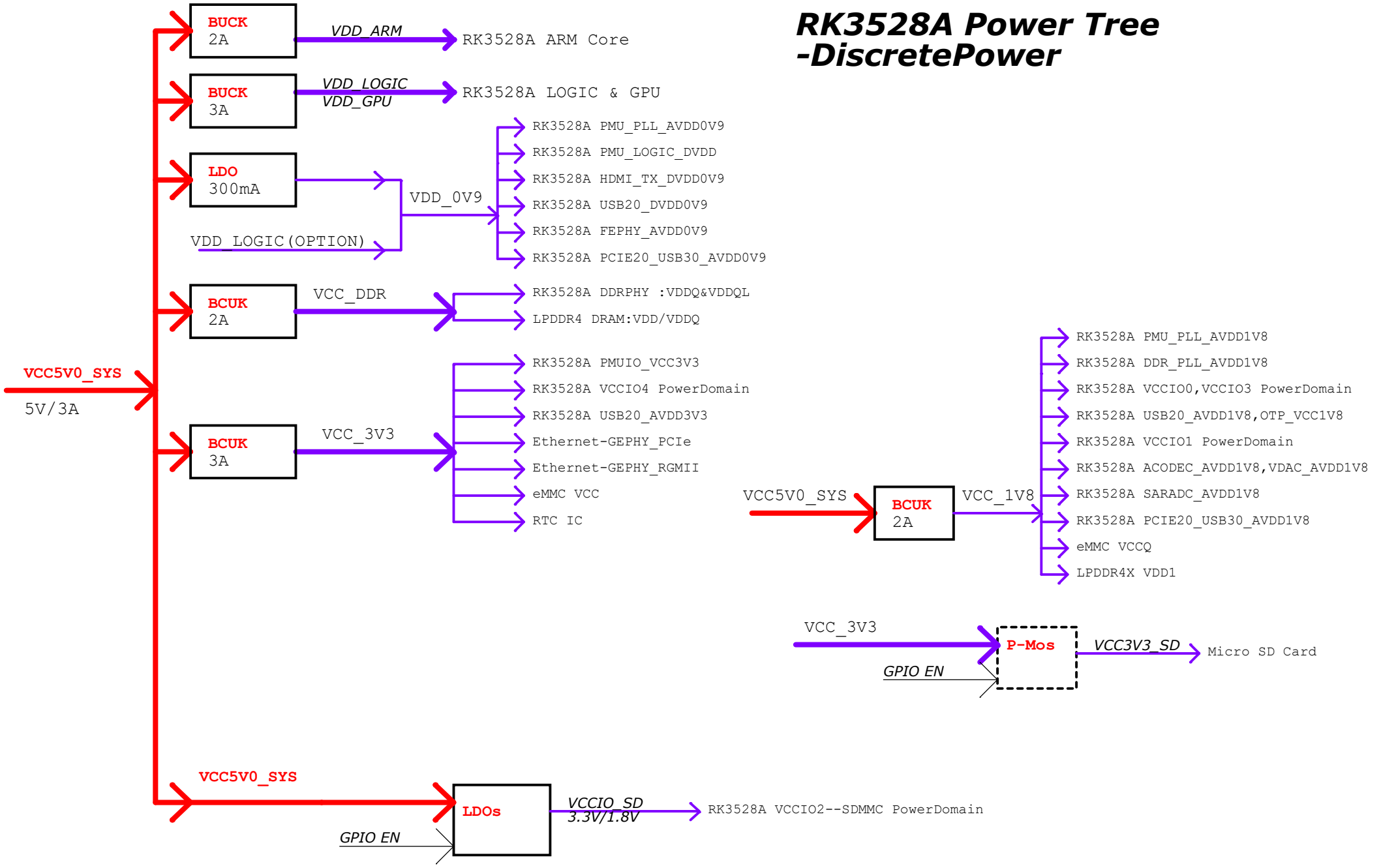
NanoPi R28S



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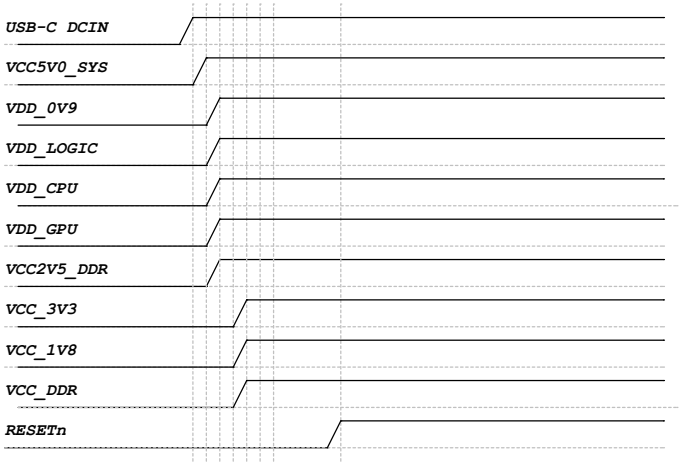
RK3528A Power Tree -DiscretePower



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DiscretePower Power Sequence



Power Supply	EXT BUCK&LDO	Supply Limit	Power Name	Time Slot	Default Voltage	Default ON/OFF	Sleep ON/OFF	Peak Current	Sleep Current
VCC5V0_SYS	BUCK1	2.0A	VDD_LOGIC VDD_GPU	Slot:1	0.9V	ON	ON	TBD	TBD
VCC5V0_SYS	BUCK2	2.0A	VDD_ARM	Slot:1	0.95V	ON	ON	TBD	TBD
VCC5V0_SYS	LDO1	0.3A	VDD_0V9	Slot:1	0.9V	ON	ON	TBD	TBD
VCC_3V3	LDO2	1.0A	VCC_1V8	Slot:2	1.8V	ON	ON	TBD	TBD
VCC5V0_SYS	BUCK3	2.0A	VCC_3V3	Slot:2	3.3V	ON	ON	TBD	TBD
VCC5V0_SYS	BUCK4	2.0A	VCC_DDR	Slot:2	ADJ FB=0.6V	ON	ON	TBD	TBD
USB-C DCIN	/	3.0A	VCC5V0_SYS	Slot:0	5.0V	ON	ON	TBD	TBD

IO Power Domain Map

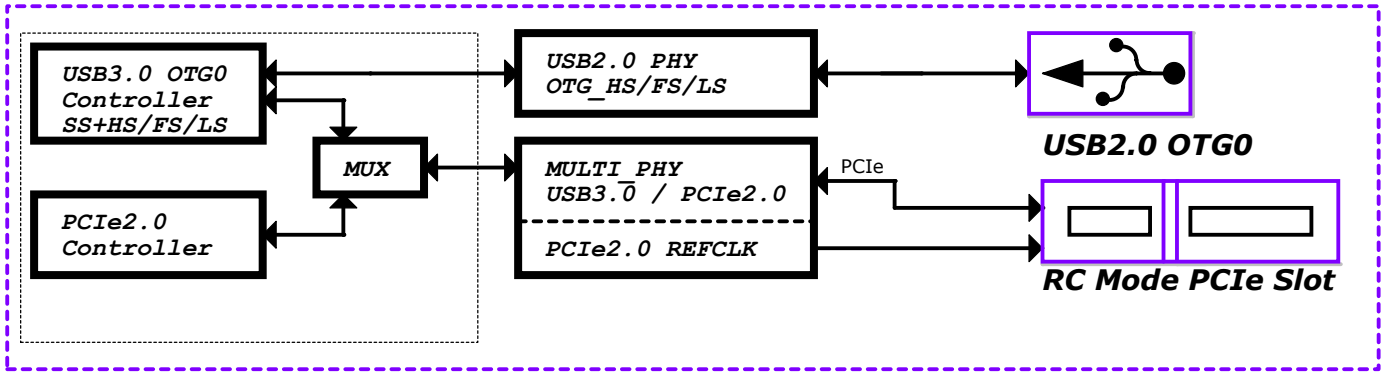
IO Domain	Pin Num	Support IO Voltage		Actual assigned IO Domain Voltage			Notes
		3.3V	1.8V	Supply Power Net Name	Power Source	Voltage	
PMUIO	Pin 1J13	✓	✗	VCCIO_PMU	VCC_3V3	3.3V	
VCCIO0	Pin 1N6	✓	✓	VCCIO0	VCCIO_WL	1.8V	
VCCIO1	Pin 1P8	✓	✓	VCCIO1	VCCIO_FLASH	1.8V/3.3V	Select as required
VCCIO2	Pin 1B14	✓	✓	VCCIO2	VCCIO_SD	1.8V/3.3V	SD3.0 Voltage=1.8V SD2.0 Voltage=3.3V 3.3V-->1.8V
VCCIO3	Pin 1M13	✓	✓	VCCIO3	VCC_1V8 VCC_3V3	1.8V 3.3V	Select as required
VCCIO4	Pin 1J3	✓	✓	VCCIO4	VCC_3V3	3.3V	

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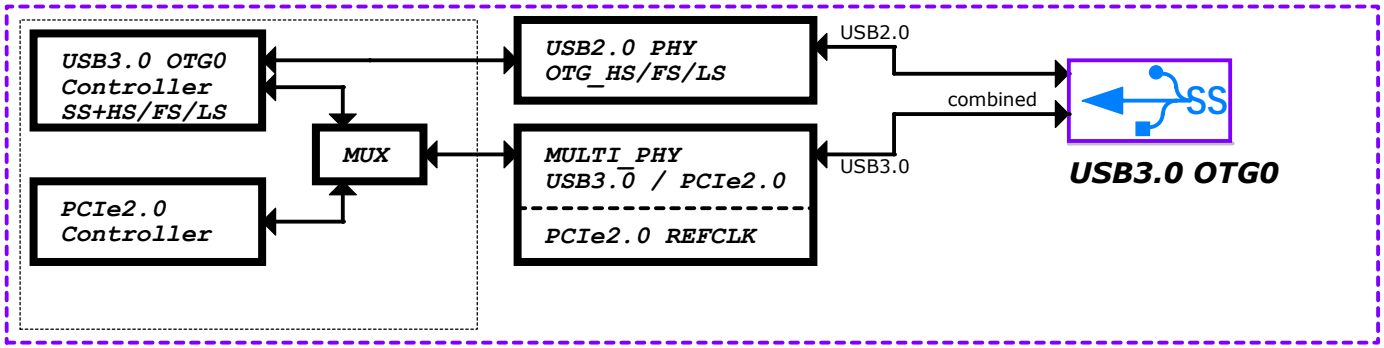
NanoPi R28S	Rev 2605
Size A3	Page Name 03.Power Sequence-DiscretePower
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USB3/PCIE Fun Map

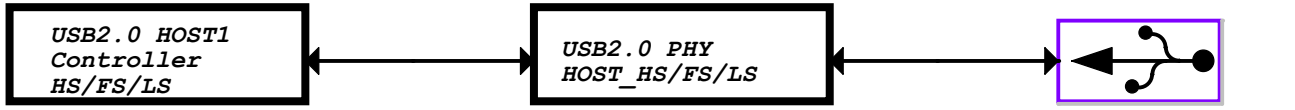
Case1: USB2.0 OTG0 + PCIe2.0 x1 Lane



Case2: USB3.0 OTG0



USB2.0 HOST1



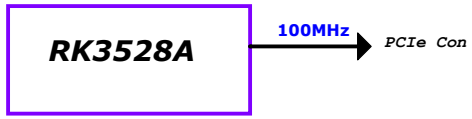
Note 1:

USB3.0 OTG is backward compatible with USB2.0, it needs to occupy the USB2.0 OTG signal pair.

Note 2:

MULTI PHY can be configured for PCIe2.0 or USB3.0 interface.
PCIe2.0 only work in RC mode,
and support both internal/external clock.

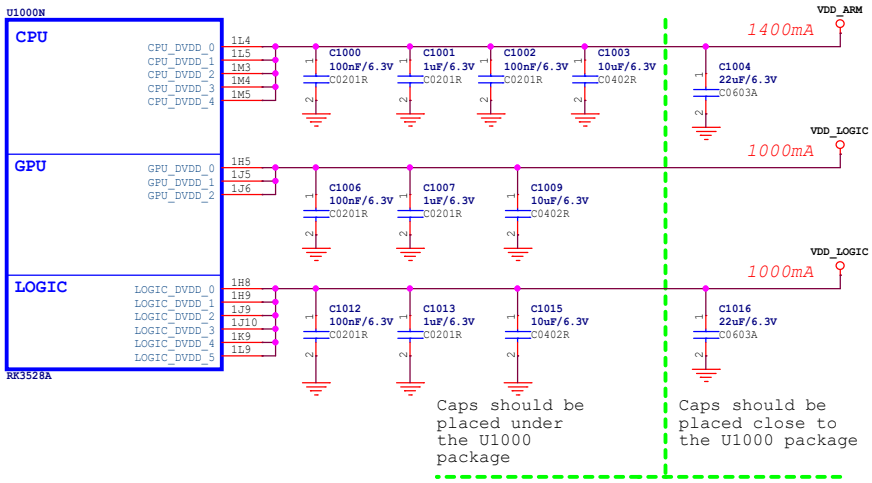
PCIe2.0 REFCLK



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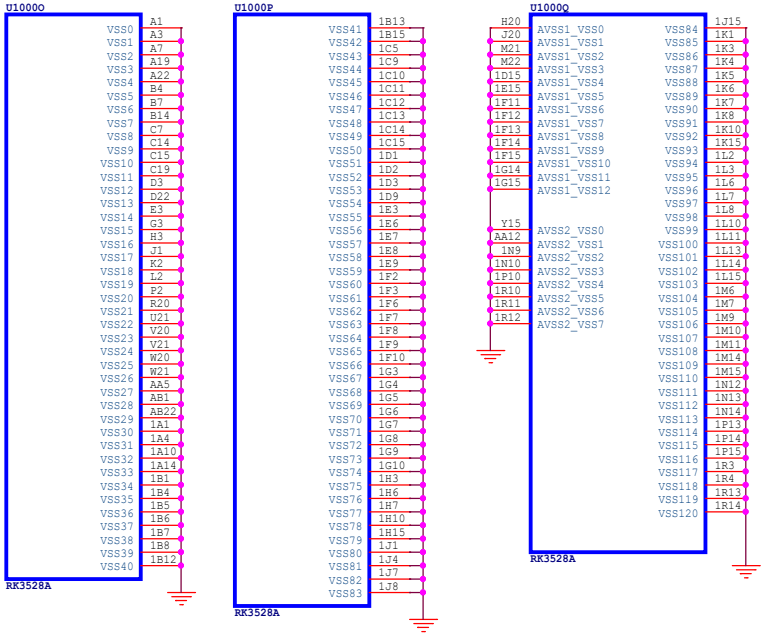
NanoPi R28S	Rev 2605
Size A3	Page Name 04.USB3/PCIE Fun Map
Date: Wednesday, July 29, 2026	Sheet: 4 / 16

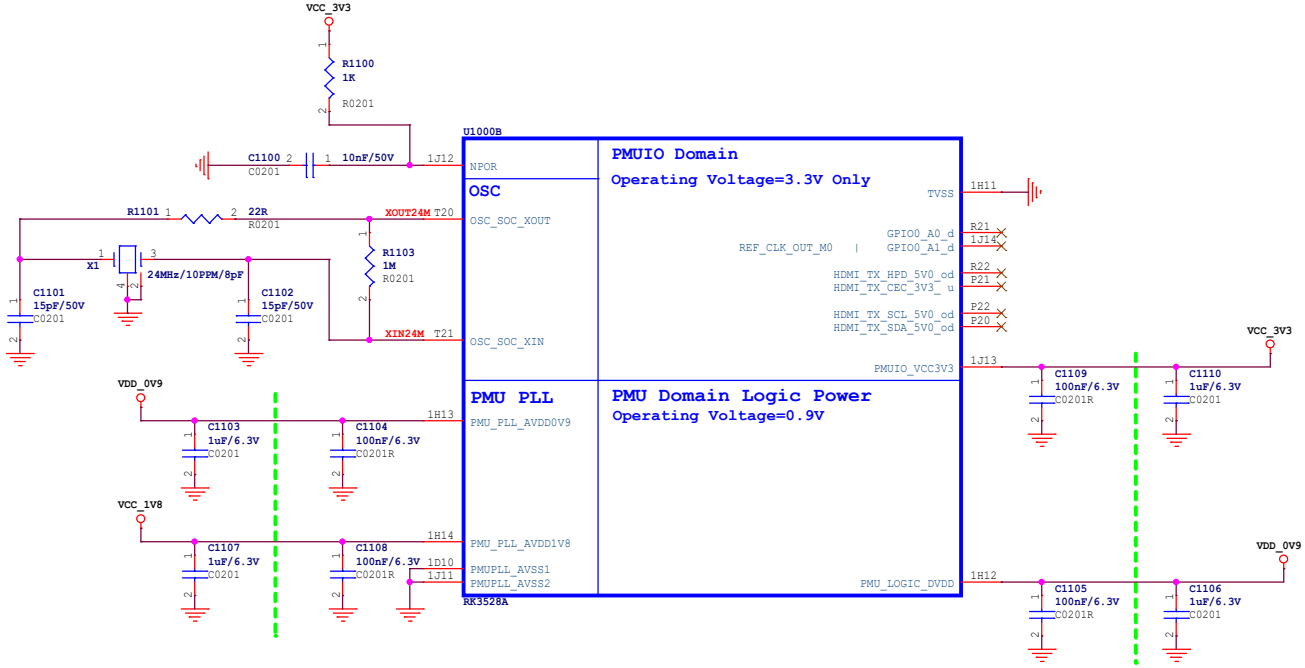
RK3528A_N(POWER)



Note:
Caps of between dashed green lines and U1000 should be placed under the U1000 package.
Other caps should be placed close to the U1000 package

RK3528A_O/P/Q(GND)



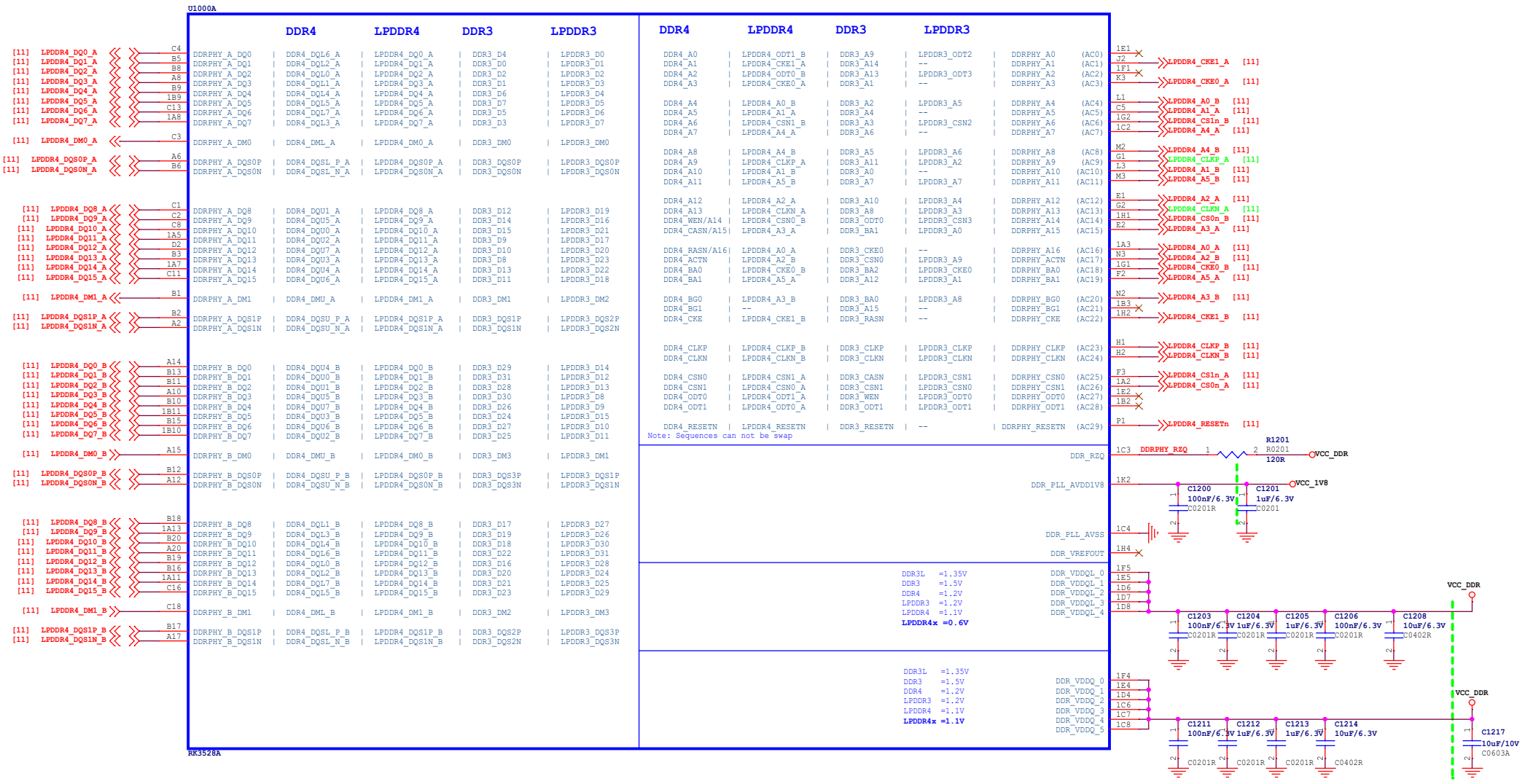


Note:
Caps of between dashed green lines and U1000 should be placed under the U1000 package.
Other caps should be placed close to the U1000 package

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RK3528A_A(DDR PHY)



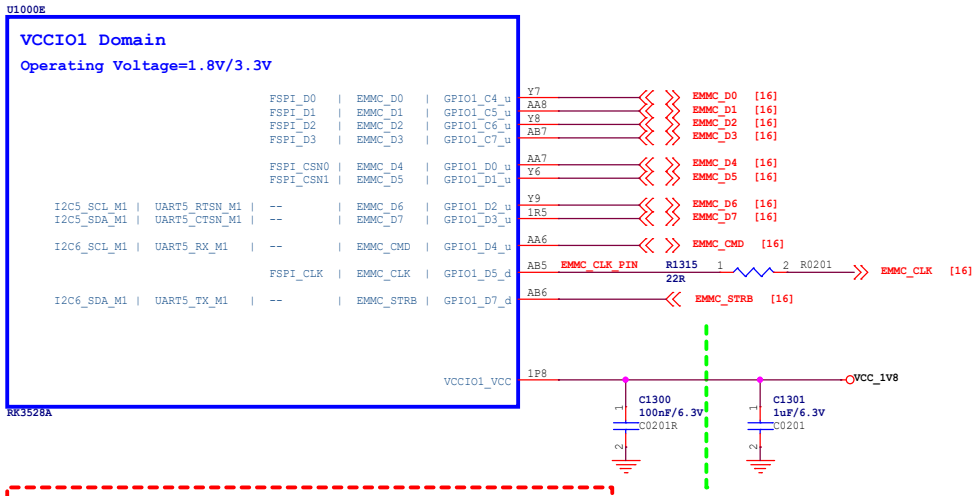
Note:

Caps of between dashed green lines and U1000 should be placed under the U1000 package.
Other caps should be placed close to the U1000 package

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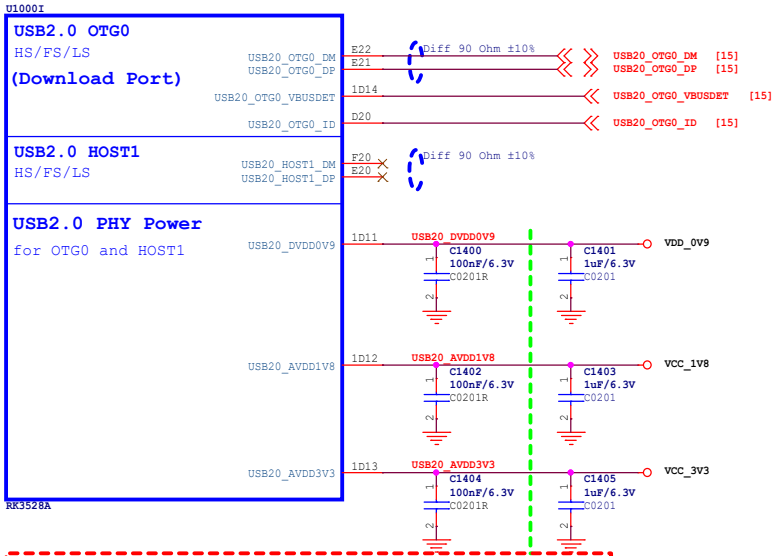
NanoPi R28S	Rev 2605
Size A3	Page Name 06.RK3528A DDR Controller
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RK3528A_E(VCCIO1 Domain)



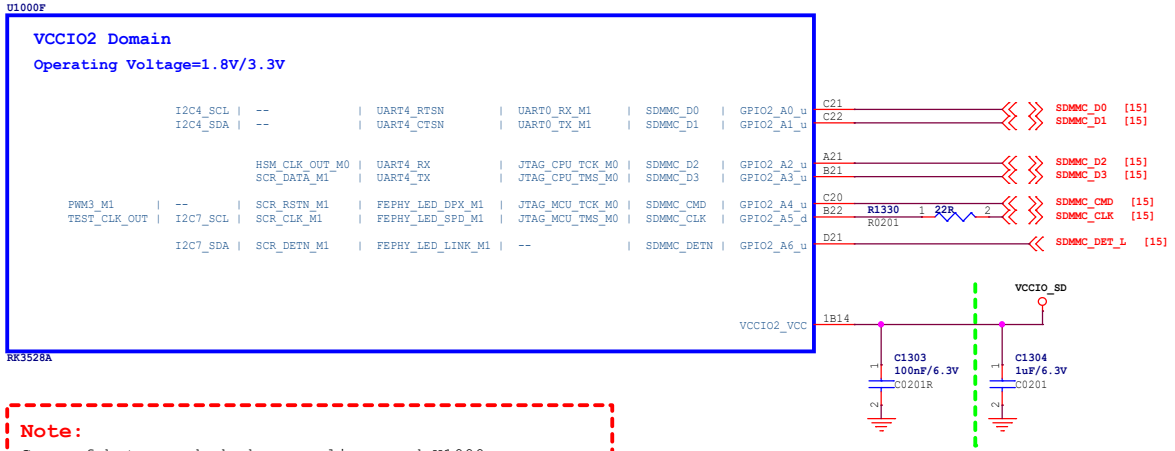
Note:
Caps of between dashed green lines and U1000
should be placed under the U1000 package.
Other caps should be placed close to the U1000 package

RK3528A_I(USB2.0 OTG/HOST)



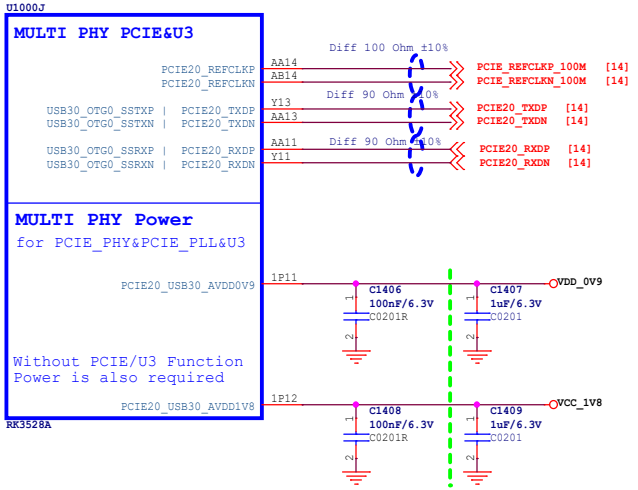
Note:
Caps of between dashed green lines and U1000
should be placed under the U1000 package.
Other caps should be placed close to the U1000 package

RK3528A_F(VCCIO2 Domain)



Note:
Caps of between dashed green lines and U1000
should be placed under the U1000 package.
Other caps should be placed close to the U1000 package

RK3528A_J(PCIE2.0/U3 PHY)

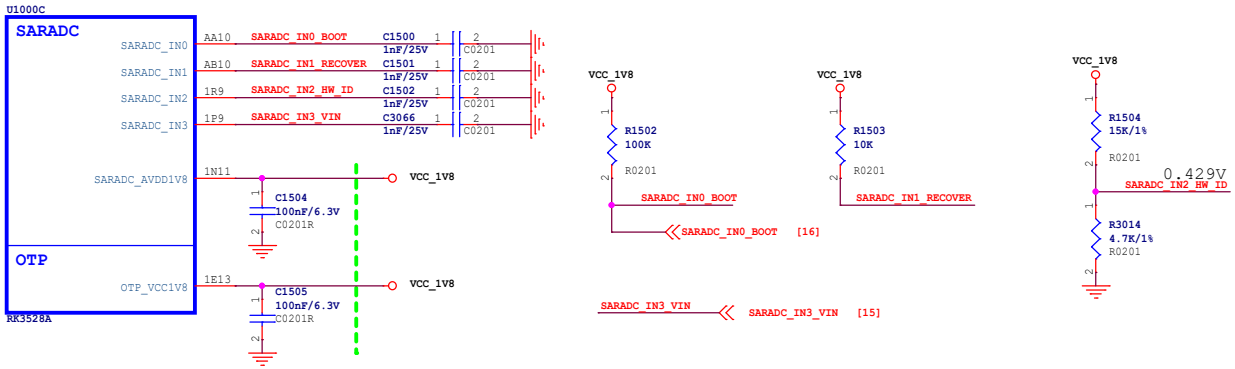


Note:
Caps of between dashed green lines
and U1000
should be placed under the U1000
package.
Other caps should be placed close to
the U1000 package

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RK3528A_C(Saradc/OTP)

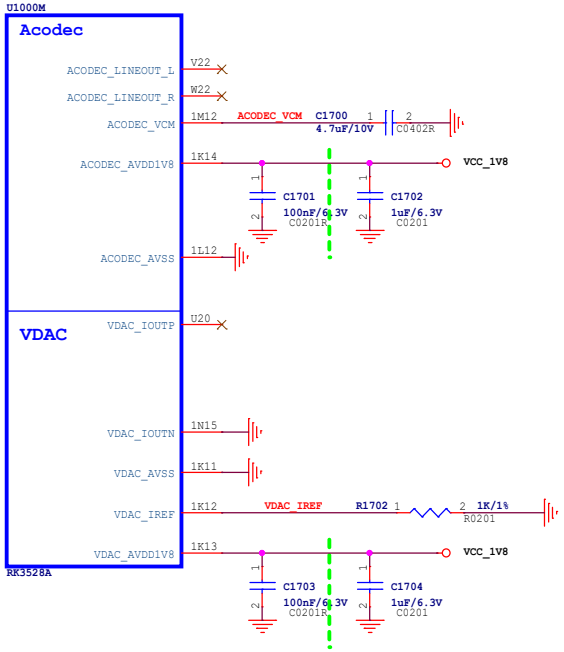


Note:
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should be placed under the U1000 package.
Other caps should be placed close to the U1000 package

TABLE

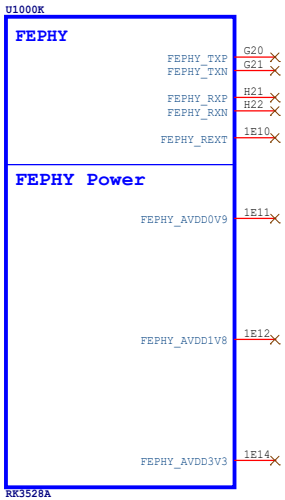
Item	Rup	Rdown	ADC	BOOT MODE
LEVEL1	*DNP*	*100K*	*0*	*USB (Maskrom mode)*
LEVEL2	*100K*	*12K*	*114*	
LEVEL3	*100K*	*27K*	*228*	*FSPI--USB*
LEVEL4	*100K*	*51K*	*342*	
LEVEL5	*100K*	*82K*	*456*	
LEVEL6	*100K*	*120K*	*570*	*EMMC--USB*
LEVEL7	*100K*	*200K*	*683*	*EMMC--SD Card--USB*
LEVEL8	*100K*	*330K*	*796*	*SD Card--USB*
LEVEL9	*100K*	*820K*	*910*	
LEVEL10	*100K*	*DNP*	*1023*	*FSPI--EMMC--SD Card--USB*

RK3528A_M(Acodec/VDAC)



Note:
Caps of between dashed green lines and U1000
should be placed under the U1000 package.
Other caps should be placed close to the U1000 package

RK3528A_K(Embed FEPHY)



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RK3528A_G(VCCIO3 Domain)

U1000G

VCCIO3 Domain

Operating Voltage=1.8V/3.3V

				TSI0_D7	UART2_RX_M0	RGMII1_TXD1	GPIO3_A0_d	AA19 → GMAC1_TXD1 [13]
				TSI0_D6	UART2_TX_M0	RGMII1_TXD0	GPIO3_A1_d	Y18 → GMAC1_TXD0 [13]
				TSI0_D5	UART2_RTSN_M0	RGMII1_RXD1	GPIO3_A2_d	Y17 → GMAC1_RXD1 [13]
				TSI0_D4	UART2_CTSN_M0	RGMII1_RXD0	GPIO3_A3_d	AA18 → GMAC1_RXD0 [13]
		UART6_CTSN	--	TSI1_D0	SDIO1_CLK	RGMII1_TXCLK	GPIO3_A4_d	AA22 GMAC1_TXCLK_PIN R1902 1 2 22R R0201 → GMAC1_TXCLK [13]
		UART6_RTSN	--	TSI1_CLKIN	SDIO1_CMD	RGMII1_RXCLK	GPIO3_A5_d	Y20 → GMAC1_RXCLK [13]
PCIE_CLKREQN_M0	--	UART6_TX_M0	--	TSI1_VALID	SDIO1_D0	RGMII1_RXD3	GPIO3_A6_d	AB21 → GMAC1_RXD3 [13]
PCIE_WAKEN_M0	--	UART6_RX_M0	--	TSI1_SYNC	SDIO1_D1	RGMII1_RXD2	GPIO3_A7_d	AA20 → GMAC1_RXD2 [13]
PCIE_PERSTN_M0	--	UART7_CTSN_M0	--	--	SDIO1_D2	RGMII1_TXD3	GPIO3_B0_d	Y19 → GMAC1_TXD3 [13]
		UART7_RTSN_M0	--	TSI0_D0	SDIO1_D3	RGMII1_TXD2	GPIO3_B1_d	AA21 → GMAC1_TXD2 [13]
		UART7_TX_M0	I2C6_SCL_M0	TSI0_CLKIN	SDIO1_PWREN	--	GPIO3_B2_d	1R15 / Y21
		UART7_RX_M0	I2C6_SDA_M0	TSI0_VALID	SDIO1_DETN	--	GPIO3_B3_d	
				TSI0_SYNC	I2S0_MCLK_M0	RGMII1_CLK	GPIO3_B4_d	Y16 → GMAC1_MCLKINOUT [13]
				TSI0_D1	I2S0_SCLK_M0	ETH_CLK_25M_OUT	GPIO3_B5_d	AB17
				TSI0_D2	I2S0_LRCK_M0	RGMII1_MDC	GPIO3_B6_d	AA17 → GMAC1_MDC [13]
				TSI0_D3	I2S0_SDI_M0	RGMII1_MDIO	GPIO3_B7_d	AA16 → GMAC1_MDIO [13]
				TSI0_FAIL	I2S0_SDO_M0	RGMII1_TXEN	GPIO3_C0_d	AB16 → GMAC1_TXEN [13]
PWM6_M2	--	UART6_TX_M1	I2C3_SCL_M1	--	--	RGMII1_TXER	GPIO3_C1_d	AA15
						RGMII1_RXDV_CRS	GPIO3_C2_d	AB19 → GMAC1_RXDV_CRS [13]
PWM5_M1	SPDIF_TX_M2	UART6_RX_M1	I2C3_SDA_M1	REF_CLK_OUT_M1	CLK_32K_OUT_M0	--	GPIO3_C3_d	AB20

VCCIO3_VCC 1M13

C1902 100nF/6.3V C0201R

C1903 1uF/6.3V C0201

VCC_1V8

RK3528A

RK3528A_H(VCCIO4 Domain)

U1000H

VCCIO4 Domain

Operating Voltage=1.8V/3.3V

PMU_DEBUG	--	I2C0_SDA_M1	--	SPDIF_TX_M0	--	--	GPIO4_A0_d	R3 → GPIO4_A0 [16]
		I2C0_SCL_M1	--	SDMMC0_PWREN	--	--	GPIO4_A1_d	R1 → SDMMC_PWREN_L [15]
		I2C1_SDA_M0	UART3_RTSN	--	--	I2S1_SDI3	GPIO4_A2_d	U2 → I2C1_SDA_M0 [15]
		I2C1_SCL_M0	UART3_CTSN	--	--	I2S1_SDI2	GPIO4_A3_d	U3 → I2C1_SCL_M0 [15]
		UART1_CTSN	--	--	PDM_CLK1	I2S1_MCLK	GPIO4_A4_d	1M2
		UART1_RTSN	--	--	--	I2S1_SCLK	GPIO4_A5_d	1P2
		UART1_TX_M0	--	--	--	I2S1_LRCK	GPIO4_A6_d	1M1
		UART1_RX_M0	--	--	--	I2S1_SDO0	GPIO4_A7_d	1P3
			UART3_RX_M0	--	--	I2S1_SDO1	GPIO4_B0_d	1R2 → LED_SYS [15]
			UART3_TX_M0	--	PDM_SDI1	I2S1_SDO2	GPIO4_B1_d	1N2 → LED1 [15]
				SPI0_MOSI	PDM_SDI0	I2S1_SDO3	GPIO4_B2_d	1N3 → GPIO4_B2_USR_BTN [16]
				SPI0_MISO	PDM_SDI2	I2S1_SDI1	GPIO4_B3_d	W3 → LED2 [15]
				SPI0_CLK	--	I2S1_SDI0	GPIO4_B4_d	R2
		FEPHY_LED_DPX_M0	--	--	PDM_CLK0	--	GPIO4_B5_d	Y1
				SPI0_CSN0	--	PWR_CTRL1	GPIO4_B6_u	1L1 → SDMMC_VOL_CTRL_H [15]
PWM4_M0	--	FEPHY_LED_SPD_M0	UART3_RX_M1	--	--	--	GPIO4_B7_d	V3
PWM5_M0	--	FEPHY_LED_LINK_M0	UART3_TX_M1	--	--	--	GPIO4_C0_d	Y2 → PCIE_ETH_PERSTB [14]
PWM6_M0	--	--	--	SPI0_CSN1	PDM_SDI3	--	GPIO4_C1_d	W2 → RTC_INT_L [15]
PWM7_M0	--	--	--	--	--	PWR_CTRL0	GPIO4_C2_d	P3 → GMAC1_RSTn_L [13]
PWM0_M0	--	--	--	GPU_AVS	--	I2C0_SDA_M0	GPIO4_C3_d	1N1
PWM1_M0	--	--	--	ARM_AVS	--	I2C0_SCL_M0	GPIO4_C4_d	V1 → PWM1_M0_ARM [12]
PWM2_M0	--	UART1_TX_M1	--	--	--	I2C1_SCL_M1	GPIO4_C5_d	1P1 → PWM2_M0_LOGIC [12]
PWM3_M0	--	UART1_RX_M1	--	--	--	I2C1_SDA_M1	GPIO4_C6_d	AA1
			UART0_RX_M0	JTAG_MCU_TMS_M1	JTAG_CPU_TMS_M1	--	GPIO4_C7_u	T3 → UART0_RX_M0_DEBUG [15]
			UART0_TX_M0	JTAG_MCU_TCK_M1	JTAG_CPU_TCK_M1	--	GPIO4_D0_d	T2 → UART0_TX_M0_DEBUG [15]

DiscretePower

I2C1_SDA_M0 R3880 1 2 2.2K R0201 VCC_3V3

I2C1_SCL_M0 R3881 1 2 2.2K R0201

VCCIO4_VCC 1J3

C1904 100nF/6.3V C0201R

C1905 1uF/6.3V C0201

VCC_3V3

RK3528A

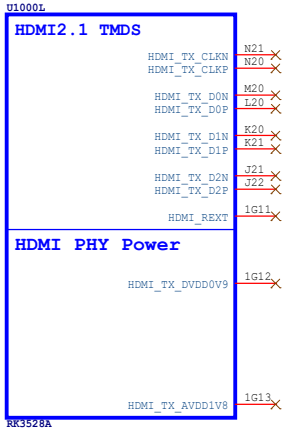
Note:

Caps of between dashed green lines and U1000 should be placed under the U1000 package.
Other caps should be placed close to the U1000 package

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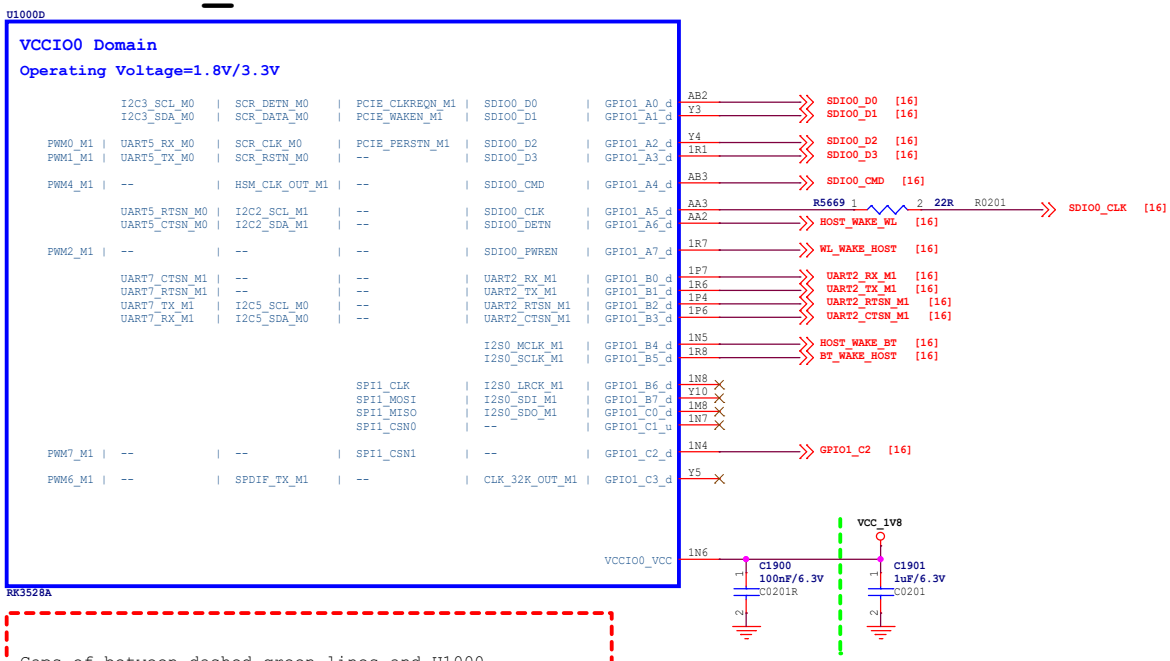
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RK3528A_L(HDMI PHY)



Note:
Caps of between dashed green lines and U1000
should be placed under the U1000 package.
Other caps should be placed close to the U1000 package

RK3528A_D(VCCIO0 Domain)

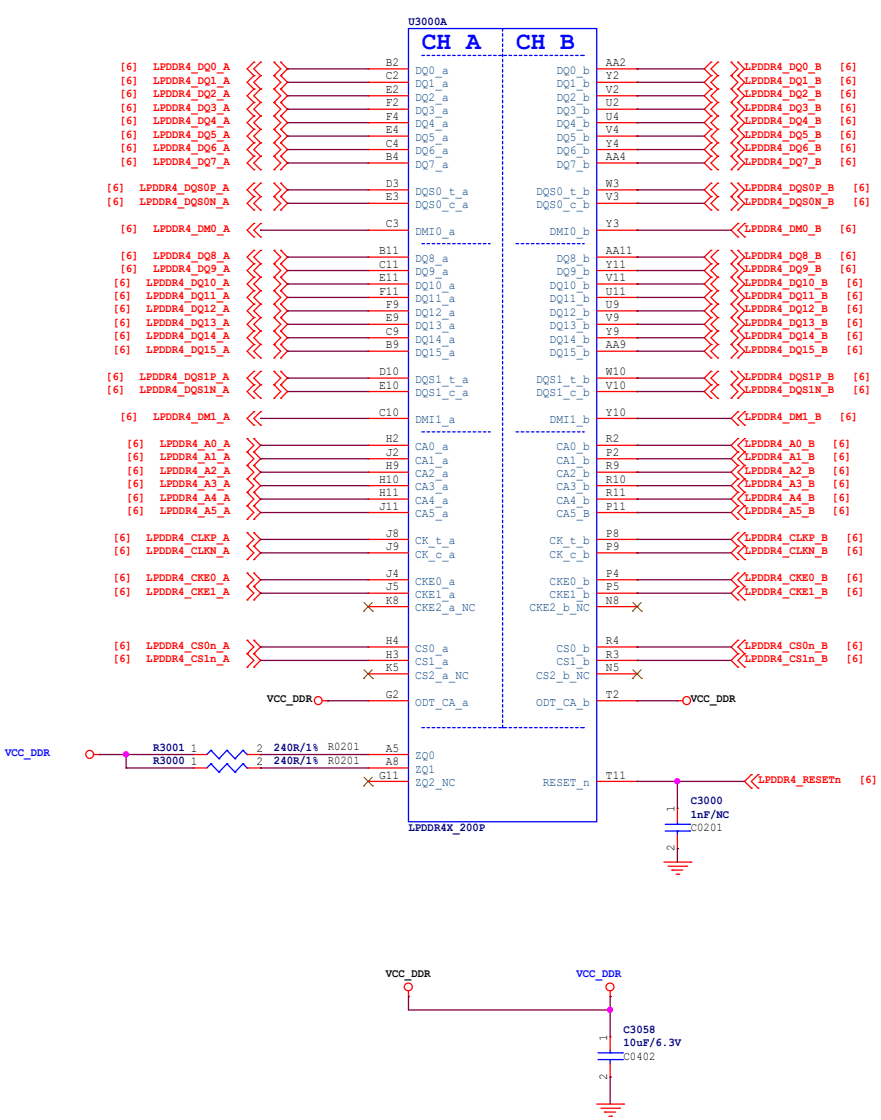


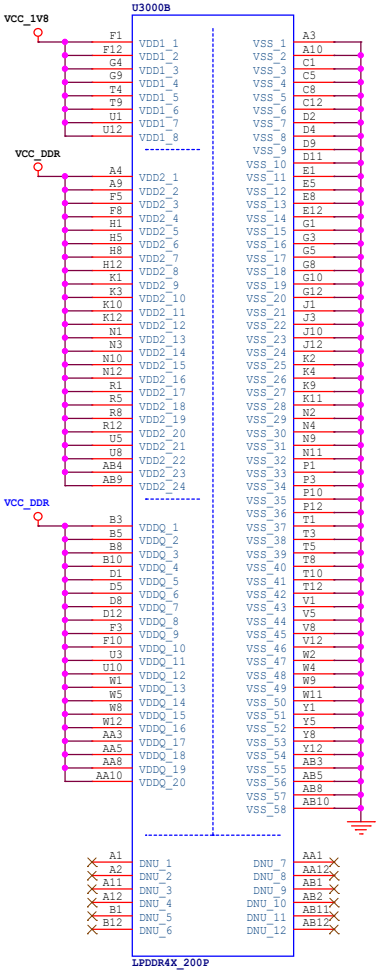
Caps of between dashed green lines and U1000
should be placed under the U1000 package.
Other caps should be placed close to the U1000 package

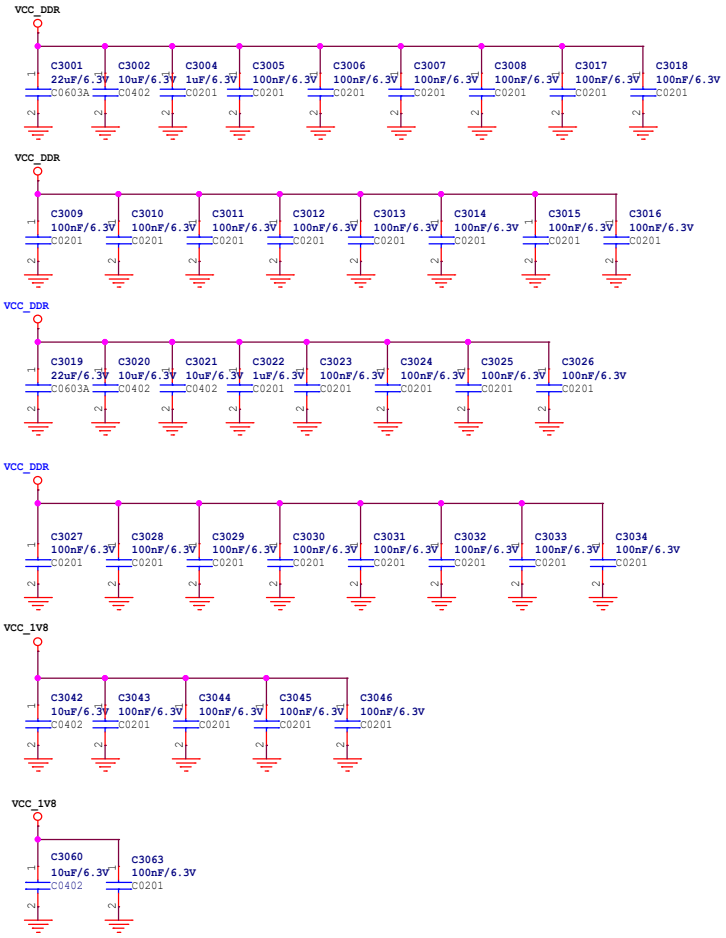
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LPDDR4
200Ball





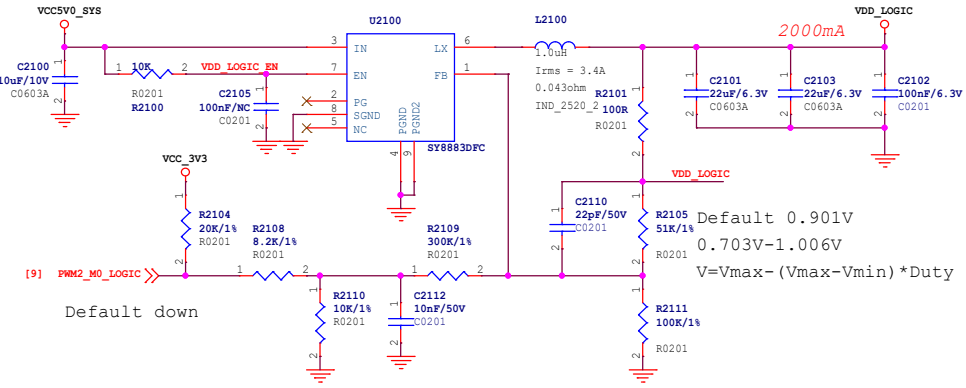


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Size A3	Page Name 11.DRAM-LPDDR4_1X32bit_200P
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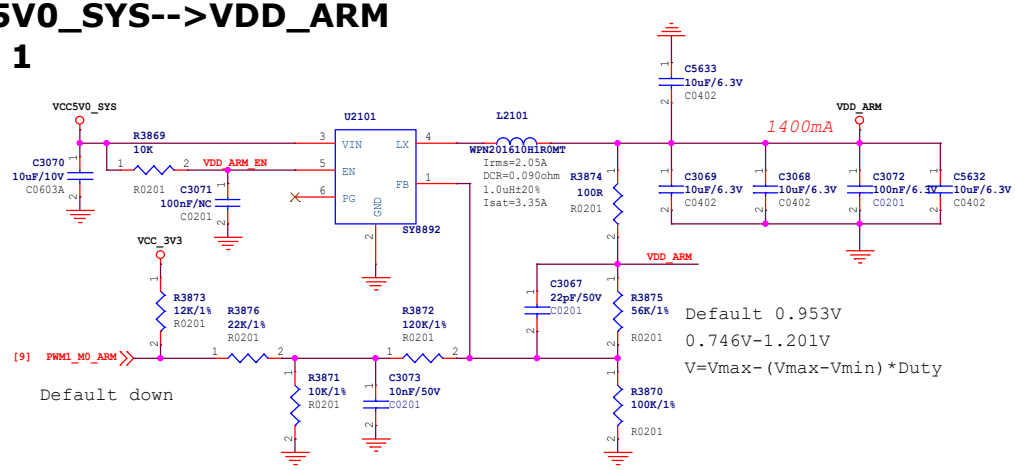
VCC5V0_SYS-->VDD_LOGIC

Step 1



VCC5V0_SYS-->VDD_ARM

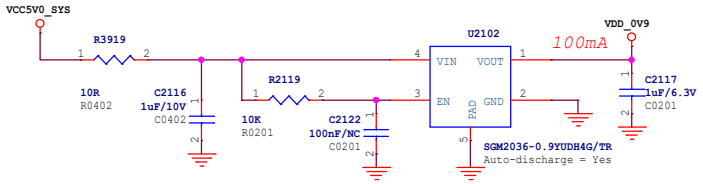
Step 1



VCC5V0_SYS-->VDD_0V9

Step 1

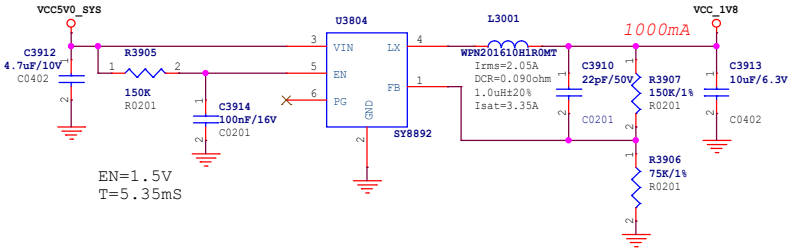
Note:
When LAYOUT, a star must be used,
Branch to each load



VCC_3V3->VCC_1V8

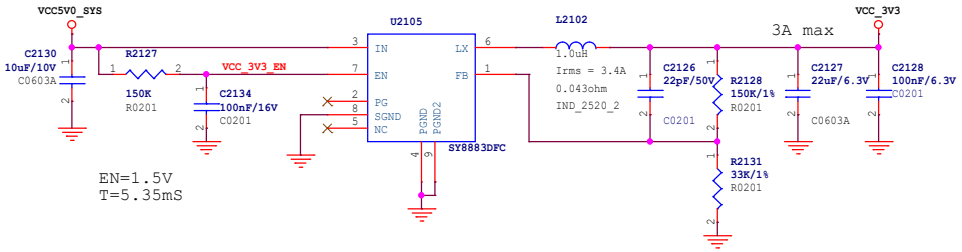
Step 2

Note:
When LAYOUT, a star must be used,
Branch to each load



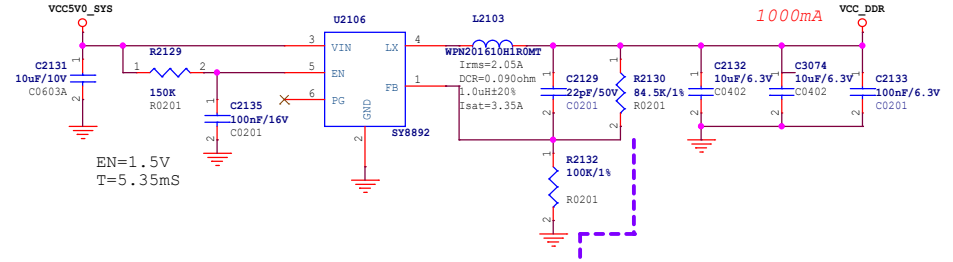
VCC5V0_SYS-->VCC_3V3

Step 2



VCC5V0_SYS-->VCC_DDR

Step 2

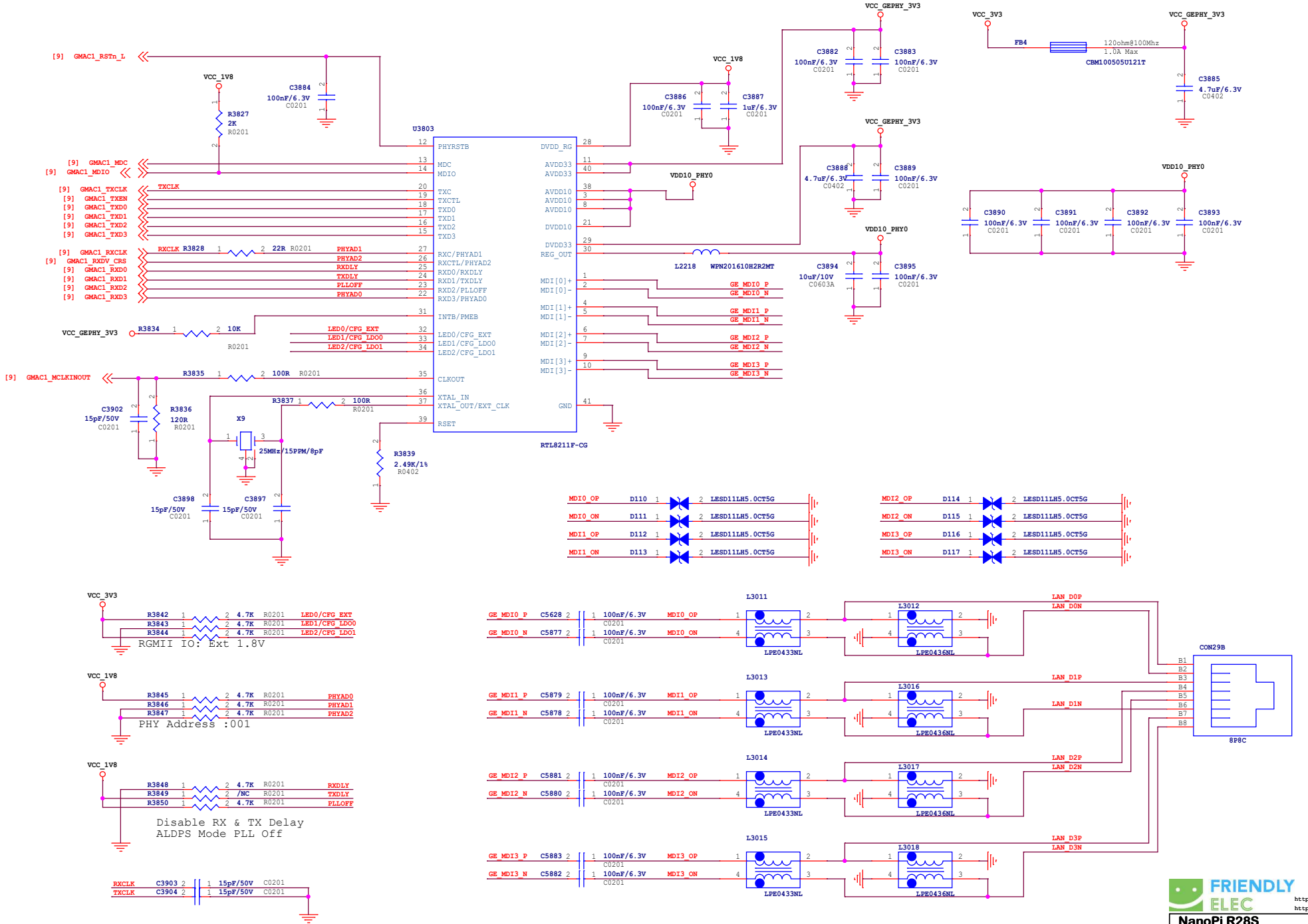


DDR4	1.20V	100K 1%
LPDDR4/4x	1.10V	84.5K 1%
DDR3	1.5V	150K 1%
DDR3L	1.35V	127K 1%
LPDDR3	1.25V	110K 1%

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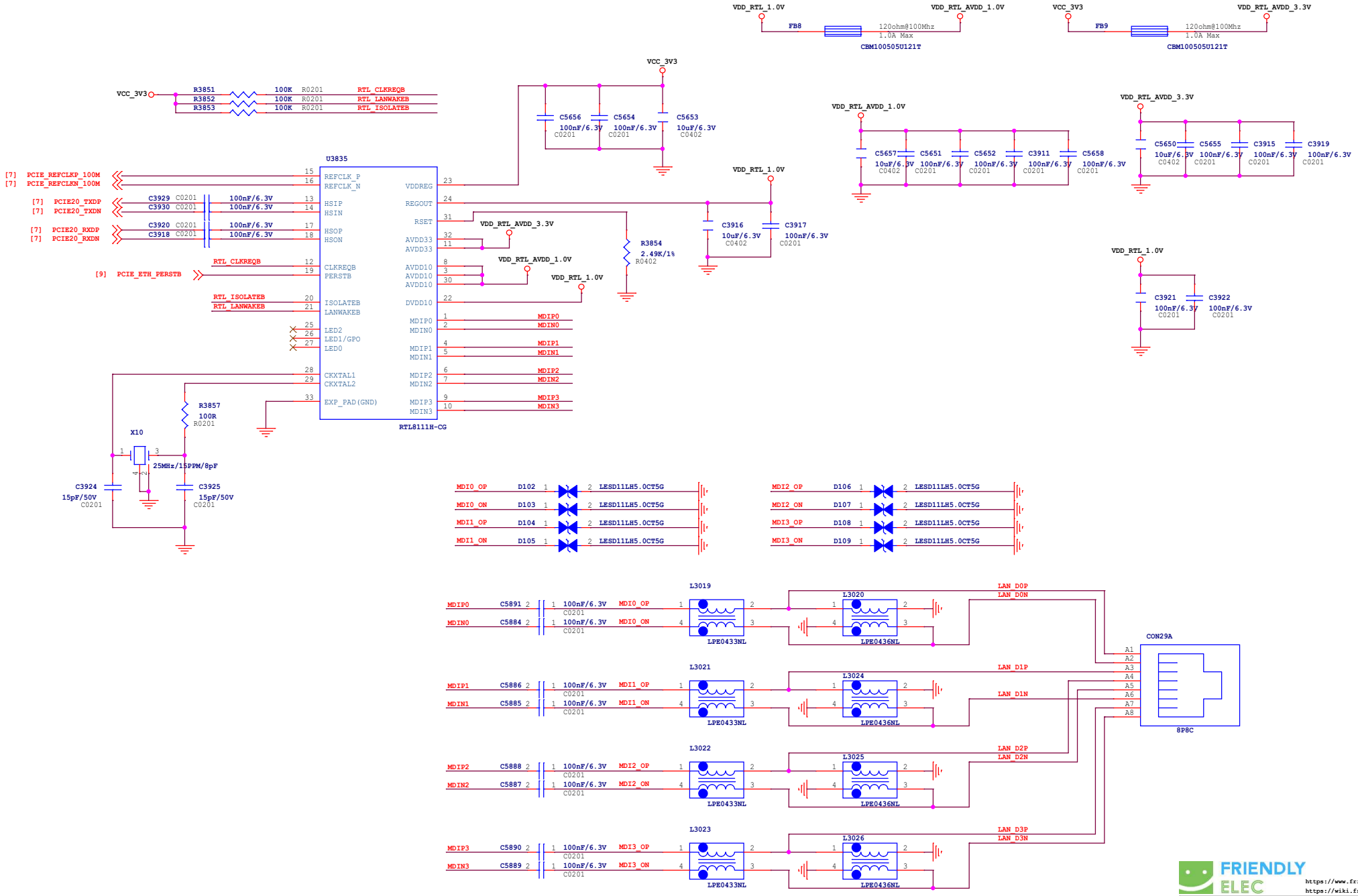
GMAC Ethernet



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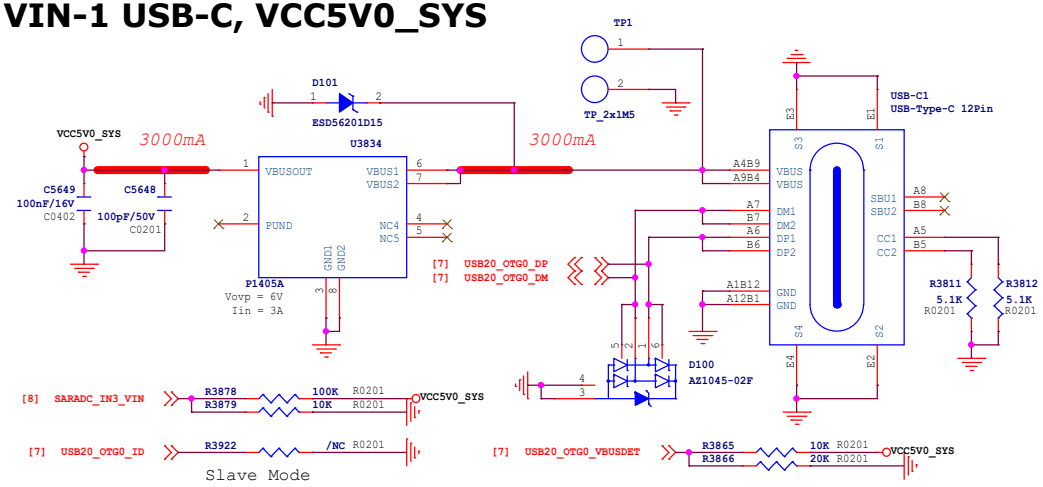
PCIe Ethernet



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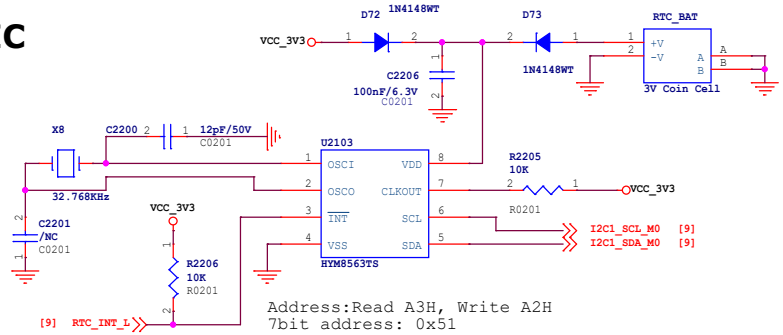
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VIN-1 USB-C, VCC5V0_SYS



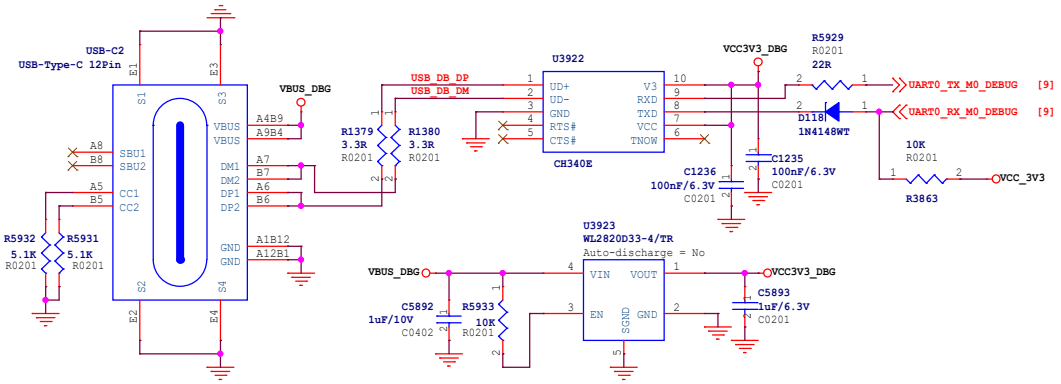
Slave Mode

RTC IC



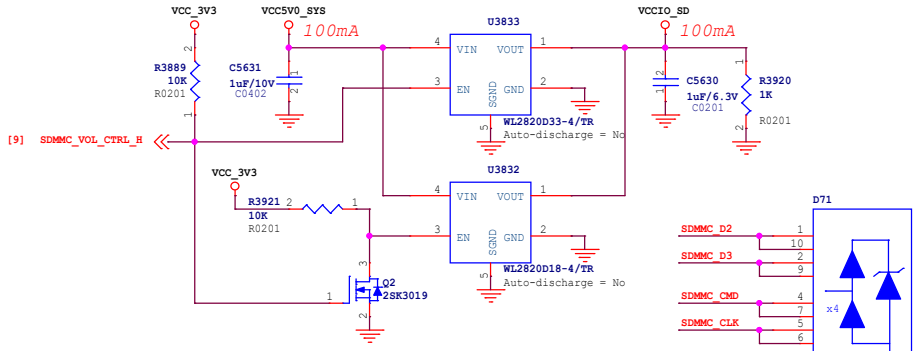
Address:Read A3H, Write A2H
7bit address: 0x51

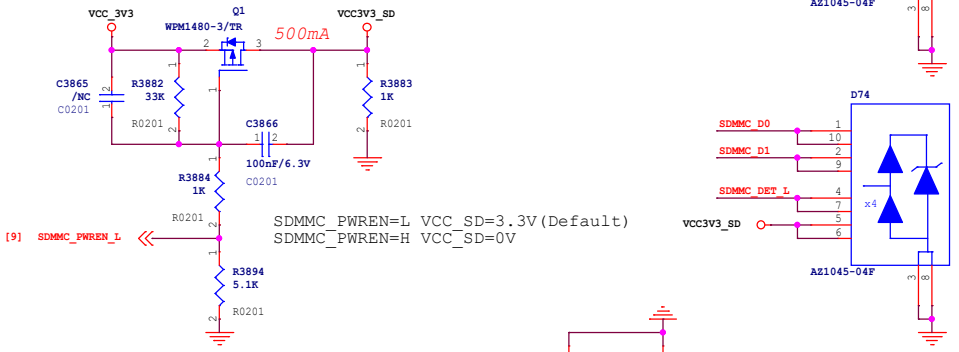
Bebug UART

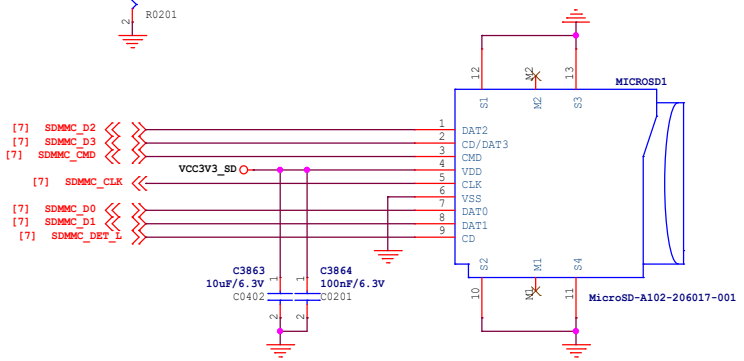


microSD

SDIO2.0 SDMMC0_VOL_CTRL=H VCCIO_SD=3.3V(Default)
SDIO3.0 SDMMC0_VOL_CTRL=L VCCIO_SD=1.8V







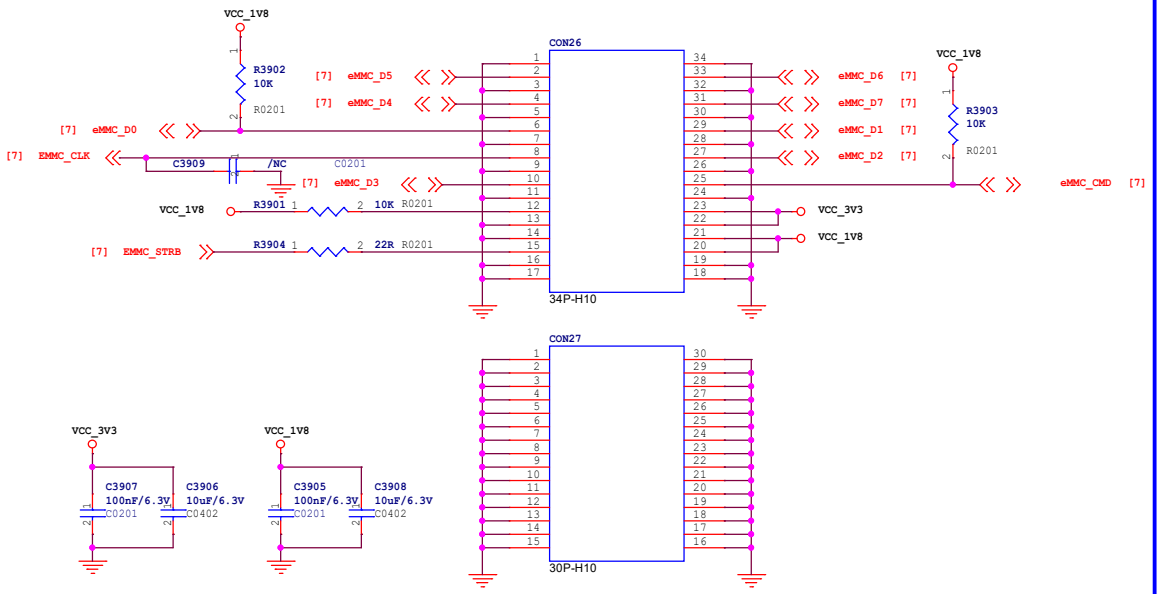
LEDs



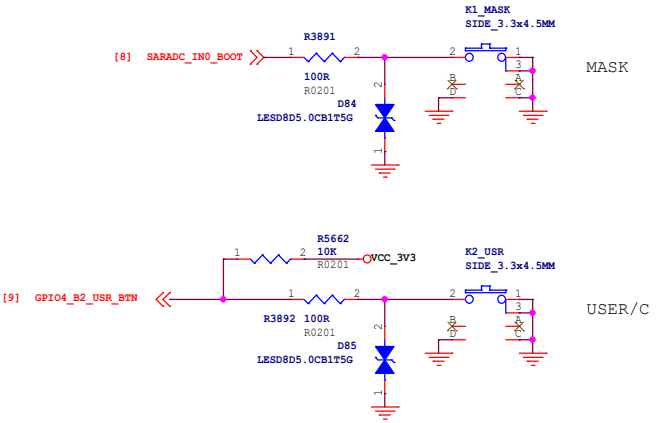
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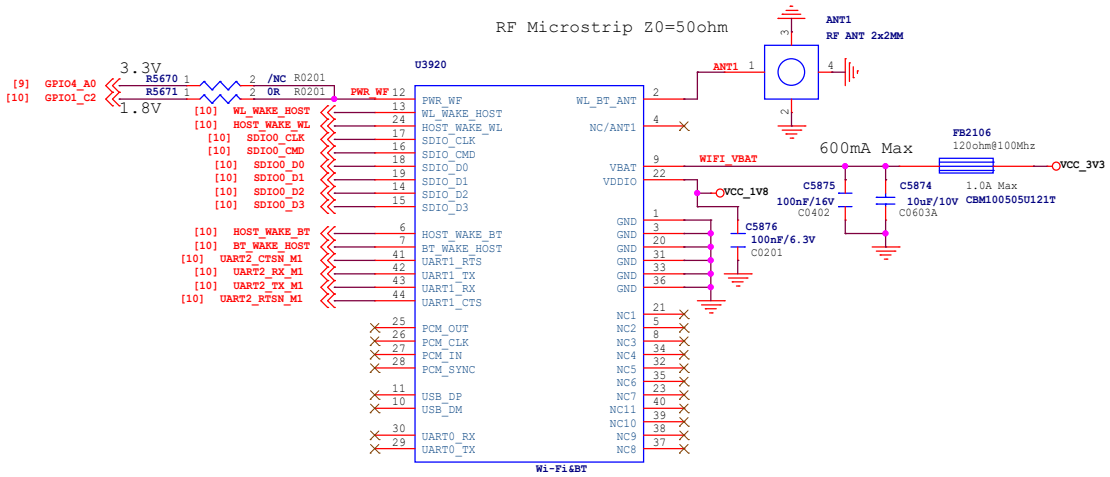
eMMC Socket



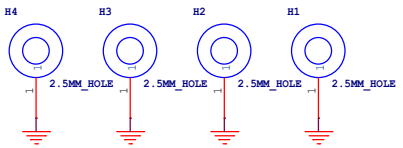
Buttons



Wi-Fi & BT



Mounting Holes



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